

Artificial Intelligence: Foundations & History

Section 1: Introduction to AI

Artificial Intelligence (AI) is a field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence. These tasks include visual perception, speech recognition, decision-making, and natural language translation.

Section 2: Machine Learning Paradigms

Machine Learning (ML) is a subset of AI where algorithms learn patterns from data rather than being explicitly programmed. The main paradigms are:

1. **Supervised Learning:** Learning from labeled data (e.g., classification, regression).
2. **Unsupervised Learning:** Finding hidden patterns in unlabeled data (e.g., clustering, PCA).
3. **Reinforcement Learning:** Learning via rewards and penalties through interaction with an environment.

Deep Learning & Transformers

Section 3: Neural Networks & Deep Learning

Deep Learning utilizes multi-layered artificial neural networks (ANNs) inspired by biological brain structures. Key architectures include Convolutional Neural Networks (CNNs) for image processing and Recurrent Neural Networks (RNNs) for sequential data.

Section 4: The Transformer Revolution

Introduced by Vaswani et al. in 2017 ('Attention Is All You Need'), the Transformer architecture replaced RNNs using self-attention mechanisms. Transformers enable parallelized processing of long text sequences and form the foundation of Large Language Models (LLMs) like GPT-4, Gemini, and Llama 3.